

Synthetic Food Colors and Neurobehavioral Hazards: The View from Environmental Health Research

Background: The proposition that synthetic food colors can induce adverse behavioral effects in children was first enunciated in 1975 by Feingold [Why Your Child Is Hyperactive. New York:Random House (1975)], who asserted that elevated sensitivity to food additives underlies the signs of hyperactivity observed in some children. Although the evidence suggested that some unknown proportion of children did respond to synthetic food colors, the U.S. Food and Drug Administration (FDA) interpreted the evidence as inconclusive. A study published in 2007 [McCann et al. Food additives and hyperactive behaviour in 3-year-old and 8/9-year-old children in the community: a randomised, double-blinded, placebo-controlled trial. Lancet 370:1560–1567 (2007)] drew renewed attention to the hypothesis because of the study's size and scope. It led the FDA to review the evidence, hold a public hearing, and seek the advice of its Food Advisory Committee. In preparation for the hearing, the FDA reviewed the available evidence and concluded that it did not warrant further agency action. **Objectives:** In this commentary I examine the basis of the FDA's position, the elements of the review that led to its decision and that of the Food Advisory Committee, and the reasons that this is an environmental health issue. **Discussion:** The FDA review confined itself, in essence, to the clinical diagnosis of hyperactivity, as did the charge to the committee, rather than asking the broader environmental question of behavioral effects in the general population; it failed to recognize the significance of vulnerable subpopulations; and it misinterpreted the meaning of effect size as a criterion of risk. The FDA's response would have benefited from adopting the viewpoints and perspectives common to environmental health research. At the same time, the food color debate offers a lesson to environmental health researchers; namely, too narrow a focus on a single outcome or criterion can be misleading.